

THE SCOTS COLLEGE

2006

TRIAL H.S.C. EXAMINATION

Mathematics

Time Allowed: 3 hours

Instructions

- ◆ Show ALL necessary working.
- ◆ Approved calculators may be used.
- ◆ All questions are of equal marks.
- ◆ **Begin each question on a new page.**

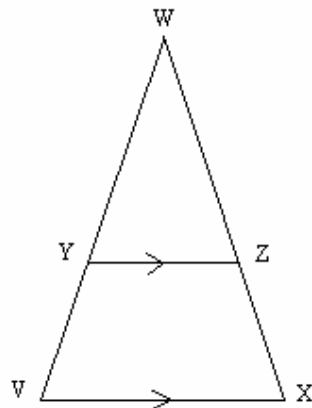
STUDENTS ARE ADVISED THAT THIS IS A TRIAL EXAMINATION ONLY AND CANNOT IN ANY WAY GUARANTEE THE CONTENT OR THE FORMAT OF THE HIGHER SCHOOL CERTIFICATE EXAMINATION

Question 1 (Begin a new Booklet)

- a) Evaluate, to 2 significant figures, $\sqrt[3]{\frac{(9.2)^2}{\pi}}$. (1)
- b) Simplify fully $\frac{6a^2 - 2ab}{9a^2 - b^2}$ (2)
- c) Solve for x : $18x^2 = 9x$ (2)
- d) Solve for x : $x^2 + x - 12 > 0$ (2)
- e) Simplify fully $e^{\ln(2x-1)}$ (1)
- f) Solve for x : $|2x - 3| \leq 10$ (2)
- g) Differentiate $\cos(3x^2)$ (2)

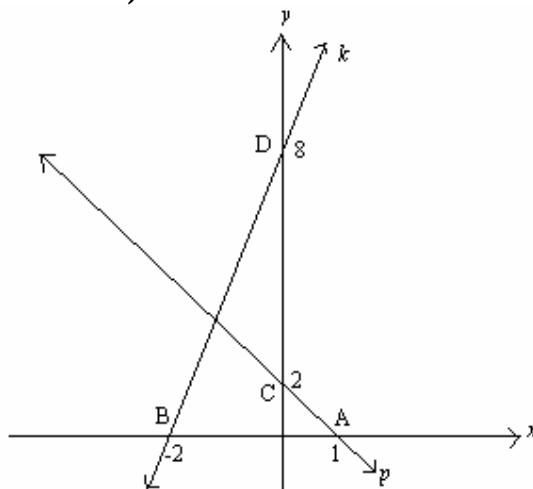
Question 2 (Begin a new Booklet)

- a) A fair coin is tossed three times. What is the probability that exactly one tail is obtained? (2)
- b) Differentiate:
- i) $x \ln x$ (2)
- ii) $\frac{2x+1}{e^x}$ (2)
- c) Solve the equation $\sqrt{5x+2} = 7$ (2)
- d) In triangle WXV , $YZ = 12$ cm, $VX = 16$ cm, $WX = 8$ cm and $YZ \parallel VX$. Prove that $\triangle WZY$ is similar to $\triangle WXV$ and find the length of WZ . (4)



NOT TO SCALE

Question 3 (Begin a new Booklet)



In the diagram above, the line p cuts the y axis at $C(0, 2)$ and the x axis at $A(1, 0)$ while the line k cuts the y axis at $D(0, 8)$ and the x axis at $B(-2, 0)$.

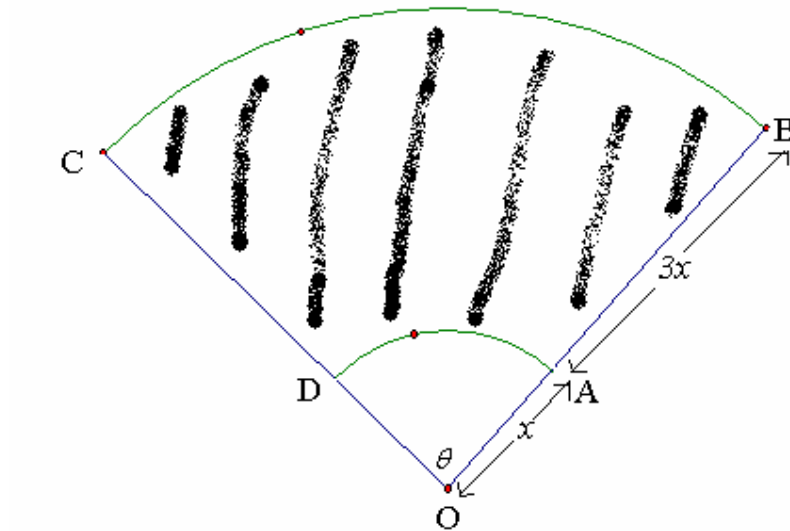
- a) Find the equation of the line k and the line p . (2)
- b) Find the coordinates of the point Q , where lines k and p intersect. (2)
- c) Write the equation of a line, q , passing through Q and perpendicular to the x axis. (2)
- d) Find the area enclosed by the lines p and q and the x axis. (2)
- e) Find the equation of the line t through the point $A(1, 0)$ and parallel to the line k . (2)
- f) Find the perpendicular distance between the line t and the line k . (2)

Question 4 (Begin a new Booklet)

- a) $P(x, y)$ moves so that its distance from $M(3, 0)$ is always twice its distance from the point $N(0, 3)$.
 - i) Show that the equation of the locus of all points $P(x, y)$ is
$$x^2 + 2x + y^2 - 8y + 9 = 0$$
(4)
 - ii) Give a geometrical description of the locus. (2)
- b) Find the equation of the curve $y = f(x)$ which satisfies the conditions:
 - $f'(x) = e^x + b$
 - $(0, 7)$ lies on the curve, and
 - the slope of the tangent at $x = 0$ is 3. (4)
- c) Find
 - i) $\int \sec^2(3x) dx$ (1)
 - ii) $\int \frac{5}{3x+2} dx$ (1)

Question 5 (Begin a new Booklet)

- a) Solve $4 - 4\cos 2x = 2$ for $0 \leq x \leq 2\pi$ (2)
- b) In the diagram below, a cars windscreen wiper blade sweeps across the region ABCD, where BC and AD are the arcs of circles with centre O. The intervals OA and AB are x cm and $3x$ cm respectively, with $\angle BOC = \theta$. The perimeter of the shaded region ABCD is 240 cm.



- i) Find the angle θ , in terms of x . Show that $\theta = \frac{240 - 6x}{5x}$ (1)
- ii) Show that the area ABCD is: $9x(40 - x) \text{ cm}^2$ (1)
- iii) Find the maximum area of the shaded region. (2)
- c) Prove $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$. (3)
- d) Differentiate $y = x \sin x + \cos x$. Hence, find in exact form, $\int_0^{\frac{\pi}{2}} x \cos x dx$. (3)

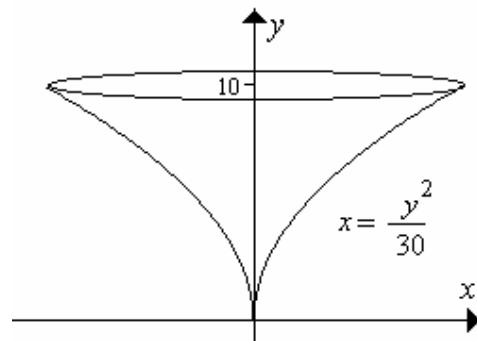
Question 6 next page....

Question 6 (Begin a new Booklet)

- a) A parabola has the equation $x^2 - 6x - 6y - 3 = 0$. For this parabola, find:
- i) the focal length,
 - ii) the coordinates of the vertex,
 - iii) the coordinates of the focus,
 - iv) the equation of the directrix (5)
- b) α and β are the roots of $(2m - 1)x^2 + (1 + m)x + 1 = 0$. Find m if $\alpha + \beta = 0$. (2)
- c) The table below shows points on a continuous curve $y = f(x)$. Use the trapezoidal rule to find the approximate value, to 2 decimal places, of $\int_3^{3.4} f(x)dx$. (2)

x	3	3.2	3.4
$f(x)$	7.19	7.62	8.41

- d) A glass shape is obtained by rotating part of the parabola $x = \frac{y^2}{30}$ about the y axis as shown. The glass is 10 cm deep.
- Find the volume of liquid, to 1 decimal place, which the glass will hold. (3)

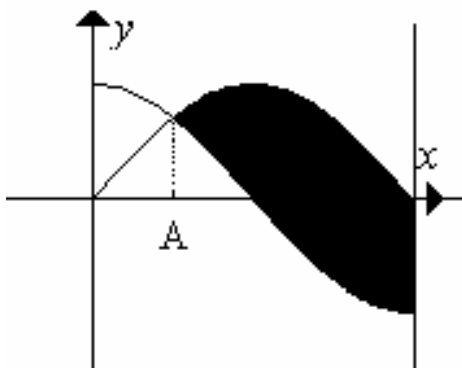


Question 7 next page....

Question 7 (Begin a new Booklet)

- a) The function $y = x^3 - 3x^2 - 9x + 1$ is defined in the domain $-4 \leq x \leq 5$.
- i) Find the coordinates of any turning points and determine their nature.
 - ii) Find the coordinates of any points of inflexion.
 - iii) Determine the minimum value of the function y in the domain $-4 \leq x \leq 5$.
- (7)

b)



The diagram above shows $y = \sin x$ and $y = \cos x$ for the domain $0 \leq x \leq \pi$.

- i) Show that the value of A, the x coordinate for the point of intersection of the two curves, is $\frac{\pi}{4}$. (2)
- ii) Find the size of the shaded area in exact form with a rational denominator in simplest form. (3)

Question 8 next page.....

Question 8 (Begin a new Booklet)

- a) The mass M of radioactive substance present after t years is given by $M = 12e^{-kt}$ where k is a positive constant. After 150 years, the mass reduced to 4 kg.
- i) What was the original mass? (1)
 - ii) Find the value of k , to 4 decimal places. (1)
 - iii) What amount of the substance would remain after a period of 500 years, to 2 significant figures? (1)
 - iv) How long, to the nearest year, would it take for the original mass to reduce to 6 kg? (2)
- b) A 12 cm bamboo plant is put into a garden and its growth is recorded. It grows 5cm every week. In what week will its height reach 167 cm? (2)
- c) The sum of the first n terms of a series is given by $S_n = 2^n + n^2$. Find the 15th term. (2)
- d) A geometric series is generated by the rule $\sum_{n=1}^8 16\left(\frac{3}{2}\right)^{n-1}$.
- i) Write out the first three terms of the series. (1)
 - ii) Find the exact value of the sum of this series. (2)

Question 9 next page.....

Question 9 (Begin a new Booklet)

- a) A closed water tank in the shape of a right cylinder is to be constructed with a surface area of $64\pi \text{ cm}^2$. The height of the cylinder is h cm and the base radius is r cm.

- i) Show that the height of the water tank, in terms of r , is given by (2)

$$h = \frac{32}{r} - r$$

- ii) Show that the volume, V , that can be contained in the tank is given by (1)

$$V = 32\pi r - \pi r^3$$

- iii) Find the radius r cm which will give the cylinder its greatest possible volume. Justify your answer. (3)

- b) Oz Challenge is a game played with two different coloured dice: one gold and the other blue.

The six faces of the blue die are numbered: 5, 7, 9, 10, 11, 13

The six faces of the gold die are numbered: 1, 4, 6, 8, 12, 14

The player wins if the number on the gold die is larger than the number of the blue die.

- i) Calculate the probability of the player winning a game. (2)

- ii) Calculate the probability that the player wins at least once in 2 successive games. (2)

- c) Solve $\log_e(x^2 - x) = \log_e 2 + \log_e(3x + 4)$ (2)

Question 10 next page.....

Question 10 (Begin a new Booklet)

- a) Daniel borrows \$8 000 from his father to pay for his World Cup Football trip and tickets. They agree that Daniel should pay interest of 1.5% every month and that he should agree to pay his father back an instalment every month.
- i) Letting \$A be the amount owing after n months, and \$T be the value of each monthly instalment, derive an expression, involving T, for the amount owing after 12 months. (1)
- ii) Hence, find the value of T, to the nearest dollar, if he repays the loan after two years. (5)
- b) A particle moves on a horizontal line so that its displacement x cm to the right of the origin at time t seconds is $x = t \sin t$.
- i) Find expressions for the velocity and acceleration of the particle. (2)
- ii) Find the exact velocity of the particle at time $t = \frac{\pi}{4}$. (1)
- iii) What effect does the acceleration have on the velocity of the particle at $t = \frac{\pi}{4}$. (1)
- iv) After the particle leaves the origin, is the particle ever at rest? Give reasons for your answer. (2)

The End

Q1

$$a) \sqrt[3]{\frac{(9.2)^2}{\pi}} = 2.9978...$$

$$= 3.0 \text{ (2 sig fig)} \quad \checkmark$$

$$b) \frac{6a^2 - 2ab}{9a^2 - b^2} = \frac{2a(3a-b)}{(3a-b)(3a+b)} \quad \checkmark$$

$$= \frac{2a}{3a+b} \quad \checkmark$$

$$c) 18x^2 = 9x$$

$$18x^2 - 9x = 0$$

$$9x(2x-1) = 0$$

$$x = 0 \quad \text{or} \quad 2x-1=0$$

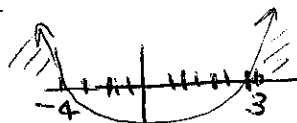
$$\checkmark \quad 2x=1 \quad \checkmark$$

$$x = \frac{1}{2} \quad \checkmark$$

$$d) x^2 + x - 12 > 0$$

$$(x+4)(x-3) > 0$$

$$x < -4 \text{ or } x > 3 \quad \checkmark$$



$$e) e^{\ln(2x-1)} = 2x-1 \quad \checkmark$$

$$f) |2x-3| \leq 10$$

$$-10 \leq 2x-3 \leq 10$$

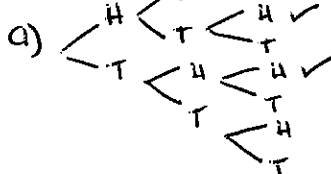
$$-7 \leq 2x \leq 13$$

$$-\frac{7}{2} \leq x \leq \frac{13}{2} \quad \checkmark$$

$$g) y = \cos(3x^2)$$

$$\frac{dy}{dx} = -6x \sin(3x^2) \quad \checkmark$$

Q2



$$P(=HT) = \frac{3}{8} \quad \checkmark$$

$$b) i) y = x \ln x$$

$$\frac{dy}{dx} = v \frac{dv}{dx} + u \frac{dv}{dx}$$

$$= \ln x \cdot 1 + x \cdot \frac{1}{x}$$

$$= \ln x + 1 \quad \checkmark \checkmark$$

$$u = x \quad v = \ln x$$

$$\frac{du}{dx} = 1 \quad \frac{dv}{dx} = \frac{1}{x}$$

$$ii) y = \frac{2x+1}{e^x}$$

$$\frac{dy}{dx} = v \frac{dv}{dx} - u \frac{dv}{dx}$$

$$= e^x \cdot 2 - (2x+1)e^x$$

$$= \frac{e^x(2-2x-1)}{(e^x)^2}$$

$$= \frac{1-2x}{e^x} \quad \checkmark \checkmark$$

$$u = 2x+1 \quad v = e^x$$

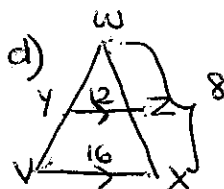
$$\frac{du}{dx} = 2 \quad \frac{dv}{dx} = e^x$$

$$c) \sqrt{5x+2} = 7$$

$$5x+2 = 49 \quad \checkmark$$

$$5x = 47$$

$$x = \frac{47}{5} \quad \checkmark$$



ΔWYZ & ΔWYX

$\angle W = \angle W$ (common) $\checkmark \checkmark \checkmark$

$\angle WYZ = \angle WYX$ (alt. $\angle$'s equal $YZ \parallel WX$)

$\angle WZY = \angle WXY$ (alt. $\angle$'s equal $YZ \parallel WX$)

$\therefore \Delta WYZ$ similar ΔWYX as they are equiangular.

$$\frac{12}{16} = \frac{x}{8}$$

$$x = 6 \text{ cm} \quad \checkmark$$

Q3.

$$a) m_k = \frac{8-0}{0-2} = 4$$

$$4 = \frac{y-8}{x}$$

$$4x = y-8$$

$$\text{"k"} \quad y = 4x+8$$

$$m_p = \frac{2-0}{0-1} = -2$$

$$-2 = \frac{y-2}{x}$$

$$-2x = y-2$$

$$\text{"p"} \quad y = -2x+2$$

$$b) 4x+8 = -2x+2 \quad y = -4+8$$

$$6x = -6 \quad y = 4$$

$$x = -1 \quad \therefore Q(-1, 4)$$

$$c) x = -1$$

$$d) \quad A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 4 \times 2$$

$$= 4 \text{ units}^2$$

$$e) m_k = 4 \quad m_t = 4 \quad \text{if parallel to k. } P(x) = 3 \text{ when } x=0$$

$$\therefore 4 = \frac{y-0}{x-1}$$

$$4x-4 = y$$

$$y = 4x-4$$

$$f) 4x-y+8=0 \quad (1,0) \quad a=4 \quad z=1$$

$$b=-1 \quad y=0$$

$$c=8$$

$$d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$$

$$= \frac{|4 \times 1 + -1 \times 0 + 8|}{\sqrt{4^2 + (-1)^2}}$$

$$= \frac{|12|}{\sqrt{17}}$$

$$= \frac{12}{\sqrt{17}} \text{ units}$$

Q4.

$$a) \quad P(x,y) \quad 2PN = PM$$

$$N(0,3) \quad M(3,0)$$

$$AN = \sqrt{(x-0)^2 + (y-3)^2}$$

$$PM = \sqrt{(x-3)^2 + (y-0)^2}$$

$$2\sqrt{x^2+(y-3)^2} = \sqrt{(x-3)^2+y^2}$$

$$4(x^2+y^2-6y+9) = (x^2-6x+9)+y^2$$

$$4x^2+4y^2-24y+36 = x^2-6x+9+y^2$$

$$3x^2+6x+3y^2-24y+27=0$$

$$(-3) \quad x^2+2x+y^2-8y+9=0$$

$$1) \quad x^2+2x+\left(\frac{2}{2}\right)^2 + y^2-8y+\left(\frac{8}{2}\right)^2 = -9+1^2+4^2$$

$$(x+1)^2 + (y-4)^2 = 8$$

$$\text{Circle centre } (-1, 4) \quad r = \sqrt{8}$$

$$= 2\sqrt{2} \text{ units}$$

$$b) f'(x) = e^x + b$$

$$e^0 + b = 3$$

$$1 + b = 3$$

$$b = 2$$

$$f'(x) = e^x + 2$$

$$f(x) = e^x + 2x + c$$

$$\text{when } x=0, f(x)=7$$

$$e^0 + 0 + c = 7$$

$$c = 6$$

$$\therefore f(x) = e^x + 2x + 6$$

$$c) \quad i) \int \sec^2 3x \, dx = \frac{1}{3} \tan 3x + c$$

$$ii) \int \frac{5}{3x+2} \, dx = \frac{5}{3} \int \frac{3}{3x+2} \, dx$$

$$= \frac{5}{3} \ln(3x+2) + c$$

Q5.

$$a) 4 - 4\cos 2x = 2$$

$$4(1 - \cos 2x) = 2$$

$$1 - \cos 2x = \frac{1}{2}$$

$$+\cos 2x = +\frac{1}{2}$$

$$2x = \frac{\pi}{3} \quad 2x = \frac{5\pi}{3} \quad 2x = \frac{7\pi}{3} \quad 2x = \frac{11\pi}{3}$$

$$x = \frac{\pi}{6} \quad x = \frac{5\pi}{6} \quad x = \frac{7\pi}{6} \quad x = \frac{11\pi}{6}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$b) i) \text{ Small } l = r\theta \quad \text{large } l = r\theta$$

$$= 3x\theta$$

$$= 4x\theta$$

$$\therefore \text{Perimeter} = 3x + 3x + x\theta + 4x\theta$$

$$240 = 6x + 5x\theta$$

$$5x\theta = 240 - 6x$$

$$\theta = \frac{240 - 6x}{5x}$$

$$ii) \text{ Area} = \frac{1}{2} \cdot 16x^2\theta - \frac{1}{2} x^2\theta$$

$$= \frac{1}{2} 15x^2\theta$$

$$= \frac{1}{2} 15x^2 \left(\frac{240 - 6x}{5x} \right)$$

$$= 3x(120 - 3x)$$

$$= 9x(40 - x)$$

$$= 360x - 9x^2 \quad \frac{dA}{dx} = 0$$

$$iii) \text{ max area occurs when } x = 20$$

$$\frac{dA}{dx} = 360 - 18x = 9 \times 20 (40 - 20)$$

$$360 - 18x = 0 = 180 (20)$$

$$18x = 360 \quad x = 20 \quad \checkmark = 3600 \text{ cm}^2$$

$$c) \frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta$$

$$= \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta = \text{RHS.}$$

$$d) y = x \sin x + \cos x$$

$$u = x \quad v = \sin x$$

$$\frac{du}{dx} = 1 \quad \frac{dv}{dx} = \cos x$$

$$\frac{dy}{dx} = \sin x + x \cos x + (-\sin x)$$

$$= x \cos x$$

$$\int x \cos x dx = [x \sin x + \cos x]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{\pi}{2} \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \right) - (0 + \cos 0)$$

$$= \frac{\pi}{2} - 1$$

Q6

$$a) x^2 - 6x - 6y - 3 = 0$$

$$(x^2 - 6x + \left(\frac{6}{2}\right)^2) = 6y + 3 + 9$$

$$(x - 3)^2 = 6(y + 2) \quad \checkmark$$

$$i) 4a = 6 \quad \text{Real length} = \frac{3}{2} \quad \checkmark$$

$$a = \frac{3}{2}$$

$$ii) \text{ vertex } (3, -2) \quad \checkmark$$

$$iii) \text{ focus } (3, -\frac{1}{2}) \quad \checkmark$$

$$iv) \text{ directrix } y = -3\frac{1}{2} \quad \checkmark$$

$$b) \alpha + \beta = -\frac{b}{a} \quad \text{but } \alpha + \beta = 0$$

$$0 = -\frac{(1+m)}{(2m-1)} \quad \checkmark \quad \text{but } m \neq \frac{1}{2}$$

$$-m - 1 = 0$$

$$m = -1 \quad \checkmark$$

$$c) A = \frac{0.2}{2} \{ 7 \cdot 19 + 8 \cdot 41 + 2 \times 7 \cdot 62 \}$$

$$= 3.084 \text{ units}^2 \quad \checkmark$$

$$d) V = \pi \int_0^b x^2 dy$$

$$= \pi \int_0^{10} \left(\frac{y^2}{30} \right)^2 dy \quad \checkmark$$

$$= \pi \int_0^{10} \frac{y^4}{900} dy$$

$$= \pi \left[\frac{y^5}{4500} \right]_0^{10} \quad \checkmark$$

$$= \pi \left[\frac{10^5}{4500} \right]$$

$$= 69.81 \dots$$

$$= 69.8 \text{ cm}^3 \quad \checkmark$$

Q7

$$a) y = x^3 - 3x^2 - 9x + 1$$

$$i) \frac{dy}{dx} = 3x^2 - 6x - 9$$

st pts occur $\frac{dy}{dx} = 0$

$$3x^2 - 6x - 9 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3 \text{ or } x = -1$$

x	-1	-	-1	-	1
$\frac{dy}{dx}$	+	0	-	0	+

$(-1, 6)$
maximum

x	3	-	3	-	3
$\frac{dy}{dx}$	-	0	+	0	+

$(3, -26)$
minimum

$$ii) \frac{d^2y}{dx^2} = 6x - 6$$

pt inflection occurs $\frac{d^2y}{dx^2} = 0$

$$6x - 6 = 0$$

$$x - 1 = 0$$

$$x = 1$$

x	1	-	1	-	1
$\frac{d^2y}{dx^2}$	-	0	+	0	+

$\therefore (1, -10)$ pt inflection

$$iii) \text{ when } x = -4 \quad y = -75$$

$$x = 5 \quad y = 6$$

$$\therefore \text{max value } -4 \leq x \leq 5$$

$$\text{is } y = -75 \text{ when } x = -4$$

$$b) i) \text{ let } x = \cos x$$

$$\frac{d \cos x}{dx} = -\sin x$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}$$

$$0 \leq x \leq \pi$$

$$ii) \int_{\frac{\pi}{4}}^{\pi} \cos x \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x \, dx + \left| \int_{\frac{\pi}{2}}^{\pi} \cos x \, dx \right|$$

$$= [-\sin x]_{\frac{\pi}{4}}^{\pi} - [-\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \left| [-\sin x]_{\frac{\pi}{2}}^{\pi} \right|$$

$$= [-\sin \pi + \sin \frac{\pi}{4}] - [-\sin \frac{\pi}{2} + \sin \frac{\pi}{4}] + \left| [-\sin \pi + \sin \frac{\pi}{2}] \right|$$

$$= (1 + \frac{1}{\sqrt{2}}) - (1 - \frac{1}{\sqrt{2}}) + |1|$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1$$

$$= 1 + \frac{2}{\sqrt{2}}$$

$$= \frac{\sqrt{2} + 2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{2 + 2\sqrt{2}}{2}$$

$$= 1 + \sqrt{2} \text{ units}^2$$

Q8

$$a) m = 12e^{-kt}$$

$$i) \text{ when } t = 0$$

$$m = 12e^0$$

$$= 12 \text{ kg}$$

$\therefore \text{initial mass} = 12 \text{ kg}$

$$ii) m = 12e^{-kt} \text{ when } m = 4, t = 150$$

$$4 = 12e^{-k \times 150}$$

$$\frac{1}{3} = e^{-150k}$$

$$\ln\left(\frac{1}{3}\right) = -150k$$

$$k = \frac{\ln\left(\frac{1}{3}\right)}{-150}$$

$$= 0.007324$$

$$= 0.0073 \text{ (to 4 dp)}$$

$$iii) m = 12e^{-0.0073t}$$

$$\text{when } t = 500$$

$$m = 12e^{-0.0073 \times 500}$$

$$= 0.31189..$$

$$= 0.31 \text{ kg (to 2 sig figs)}$$

$$iv) m = 12e^{-0.0073t}$$

$$\text{when } m = 6$$

$$6 = 12e^{-0.0073t}$$

$$\frac{1}{2} = e^{-0.0073t}$$

$$\ln\left(\frac{1}{2}\right) = -0.0073t$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-0.0073}$$

$$= 94.95$$

$$= 95 \text{ yrs. (nearest yr)}$$

$$b) 12 + 17 + 22 + \dots + 167 \quad a = 12, d = 5, T_n = 167$$

$$T_n = a + (n-1)d$$

$$167 = 12 + (n-1)5$$

$$155 = 5n - 5$$

$$5n = 160$$

$$n = 32$$

$\therefore 32\text{nd week}$

$$c) S_n = 2^n + n^2$$

$$S_{15} = 2^{15} + 15^2$$

$$= 32768 + 225$$

$$= 32993$$

$$S_{15} = 2^{15} + 15^2 \quad T_{15} = S_{15} - S_{14}$$

$$= 16580 = 32993 - 16413$$

$$\frac{1}{2}$$

$$T_{15} = 16413$$

$$d) \sum_{n=1}^8 16 \left(\frac{3}{2}\right)^{n-1}$$

$$i) n = 1$$

$$n = 1 \quad 16 \left(\frac{3}{2}\right)^0 = 16$$

$$n = 2 \quad 16 \left(\frac{3}{2}\right)^1 = 24$$

$$n = 3 \quad 16 \left(\frac{3}{2}\right)^2 = 36$$

$$\text{1st term } a = 16$$

$$\text{2nd term} = 24$$

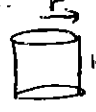
$$\text{3rd term} = 36$$

$$ii) S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{16 \left(\left(\frac{3}{2}\right)^8 - 1\right)}{\frac{3}{2} - 1}$$

$$= 32 \left(\frac{6561}{256} - 1\right)$$

$$= 788 \frac{1}{8}$$

Q9.  SA = $2\pi r^2 + 2\pi rh$
 a) $2\pi r^2 + 2\pi rh = 64\pi$ ✓
 $2\pi r(r+h) = 64\pi$
 $r(r+h) = 32$
 $r+h = \frac{32}{r}$
 $h = \frac{32}{r} - r$ ✓

b) $V = \pi r^2 h$
 $= \pi r^2 (\frac{32}{r} - r)$
 $= 32\pi r - \pi r^3$ ✓

iii) $\frac{dV}{dr} = 32\pi - 3\pi r^2$
 max volume occurs $\frac{dV}{dr} = 0$ ✓
 $32\pi - 3\pi r^2 = 0$
 $3\pi r^2 = 32\pi$
 $r^2 = \frac{32\pi}{3\pi}$
 $= \frac{32}{3}$ ✓
 $r = \pm \sqrt{\frac{32}{3}}$ cm
 as $r > 0$ $r = \sqrt{\frac{32}{3}}$ cm
 $\frac{dy}{dx} \begin{array}{c|c|c|c} \sqrt{\frac{32}{3}} & - & \sqrt{\frac{32}{3}} & \sqrt{\frac{32}{3}} \\ \hline + & 0 & - & \end{array}$
 ✓

$\therefore$ max. value. $T = \frac{8000 \times 1.015^{24}}{28.63...}$ ✓

Blue Gold

b) 5	6
	8
	12
	14
7	10
	14
9	12
	14
10	12
	14
11	12
	14
13	14

$n(\text{win}) = 14$ ✓
 i) $P(\text{win}) = \frac{14}{36}$
 $= \frac{7}{18}$ ✓
 $P(\text{loss}) = \frac{11}{18}$
 ii) $P(\text{at least once})$
 $= 1 - P(2L)$
 $= 1 - (\frac{11}{18} \times \frac{11}{18})$
 $= 1 - \frac{121}{324}$
 $= \frac{203}{324}$

c) $\log_e(x^2-x) = \log_e 2 + \log_e(3x+4)$
 $\log_e(x^2-x) = \log_e 2(3x+4)$
 $x^2-x = 6x+8$ ✓
 $x^2-7x-8=0$
 $(x-8)(x+1)=0$
 $x-8=0$ or $x+1=0$
 $x=8$ or $x=-1$ ✓

Q10

a) 1st Mth = $8000 \times 1.015 - T$
 1) 2nd Mth = $(8000 \times 1.015 - T) \times 1.015 - T$
 $= 8000 \times 1.015 \times 1.015 - T \times 1.015 - T$
 3rd Mth = $[2\text{nd Mth}] \times 1.015 - T$
 $= 8000 \times 1.015^3 - 1.015^2 T - 1.015 T - T$

$A_{24} = 8000 \times 1.015^{24} - 1.015^{23} T - \dots - 1.015 T - T$ ✓

ii) $A_{24} = 8000 \times 1.015^{24} - 1.015^{23} T - \dots - 1.015 T - T$
 $= 8000 \times 1.015^{24} - T(1.015^{23} + \dots + 1.015 + 1)$
 But $A_{24} = 0$

$8000 \times 1.015^{24} = T(1 + 1.015 + \dots + 1.015^{23})$
 $T = \frac{8000 \times 1.015^{24}}{1 + 1.015 + \dots + 1.015^{23}}$ ✓

GP $a=1, r=1.015, n=24$ ✓

$S_n = \frac{a(r^n - 1)}{r - 1}$
 $= \frac{1(1.015^{24} - 1)}{1.015 - 1}$
 $= 28.63...$ ✓

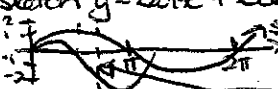
$T = \frac{8000 \times 1.015^{24}}{28.63...}$ ✓
 $= \$399.39$ every month ✓

b) $x = t \text{ ant}$
 i) $\frac{dx}{dt} = v = \text{ant} + t \text{ cost}$ ✓ $u=t, v=\text{ant}$
 $\frac{dv}{dt} = a = \text{cost} + \text{cost} - t \text{ ant}$
 $a = 2\text{cost} - t \text{ ant}$ ✓ $u=t, v=\text{cost}$
 $\frac{dv}{dt} = 1, \frac{dv}{dt} = -\text{ant}$

ii) when $t = \frac{\pi}{4}$ $v = \text{ant} \frac{\pi}{4} + \frac{\pi}{4} \text{cost} \frac{\pi}{4}$
 $= \frac{1}{\sqrt{2}} + \frac{\pi}{4} \times \frac{1}{\sqrt{2}}$ ✓
 $= \frac{1}{\sqrt{2}} + \frac{\pi}{4\sqrt{2}}$ cm/s

iii) $a = 2 \text{cost} \frac{\pi}{4} - \frac{\pi}{4} \text{ant} \frac{\pi}{4}$
 $= 2 \times \frac{1}{\sqrt{2}} - \frac{\pi}{4} \times \frac{1}{\sqrt{2}}$
 $= \frac{2}{\sqrt{2}} - \frac{\pi}{4\sqrt{2}}$ cm/s² ✓

as acceleration is positive, the particle is speeding up the particle.

iv) at rest $v=0$ i.e. $\text{ant} + t \text{ cost} = 0$ ✓
 sketch $y = \text{ant} + t \text{ cost}$

 we can see that it will be at rest again at this point the y values added = 0